

Written Status Report

CMSI 4072 Project Status Sheet Spring 2026

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Project: Sammy's Games: A Board and Card Game Platform

Date: February 27, 2026

Period Covered: January 12 - February 27

## **Accomplishments since last report:**

Designed and implemented a 4x4 grid board game using HTML, CSS, and JavaScript. The board generates cells and tracks game state through a single boardState array.

Implemented the rules and mechanics for my game *Knockout*, including:

- Orthogonal movement with obstruction checking
- Adjacent push mechanics
- Multi-piece pushing
- Edge detection and piece removal
- Win detection when a piece is pushed off the board

Developed a move-validation system (validMove) that enforces:

- Row/column movement restrictions
- No jumping over occupied spaces
- Not allowing push in same direction as previous turn

Implemented turn management and move tracking system, displaying last move direction for each player and enforcing rule constraints accordingly.

Designed and implemented an AI-controlled bot opponent featuring:

- Full legal move generation
- Board-state simulation via cloned arrays
- Detection and execution of immediate winning moves
- Avoidance of self-eliminating moves
- Randomized selection among safe moves

Created a separate simulateMove function to evaluate hypothetical moves without mutating live game state.

Added visual feedback to improve usability:

- Highlighting selected pieces
- Highlighting valid movement squares
- Visual move-direction tracking

Conducted informal user testing with 3 peer testers. Testing focused on:

- Rule clarity
- Edge-case hunting
- General usability and interface clarity

Feedback resulted in improvements to visual highlighting and clarification of push-direction rules.

### **Scheduled tasks to be done by next report:**

Implement game over lock to prevent interaction after the win condition is met (March 6).

Improve AI decision logic to detect when it is in danger of losing and act on it (March 6).

Add restart/reset functionality to support repeated structured testing sessions (March 6).

Begin planning for second game implementation, a solitaire card game called 6 Hands (March 11).

### **Noteworthy risks, concerns, or problems:**

#### **Risk: Timeline for additional games.**

The semester is approximately halfway complete, and the project goal includes 2–3 fully polished games. Developing additional games with computer opponents may require more time than anticipated.

Mitigation Plan: Focus on making the knockout bot strong, not so much other computer opponents. The second game is already set to not require an AI opponent.

#### **Concern: Never ending game.**

As of now, it is possible for a game of Knockout to last forever.

Mitigation Plan: Add detection of threefold repetition, a rule that calls the game a draw if the exact same position has been reached 3 times.